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EXPANDABLE INTRODUCER APPARATUS AND METHODS OF GUIDING AN OBJECT

Abstract

An expandable introducer apparatus comprising an inner tubular layer comprising an inner lumen and a radially expandable region extending along an elongated axis of the inner lumen and configured to at least partially radially expand relative to the elongated axis from a contracted orientation to an expanded orientation. An outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer. A flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer, and the flexible region is not bonded to the radially expandable region of the inner tubular layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Patent Application Ser. No. 63/559,974, filed Mar. 1, 2024, the entire content of which is incorporated herein by reference.

FIELD

[0002] The present disclosure relates generally to an expandable introducer apparatus and methods of guiding an object, and more particularly to expandable introducer apparatus configured to introduce an object into the vasculature of a patient and methods of guiding an object through an expandable introducer apparatus.

BACKGROUND

[0003] During medical procedures, for example, a transcatheter aortic valve replacement, a plurality of objects are commonly introduced into a patient's vasculature. Such objects may include transcatheter valve prosthesis delivery systems. These delivery systems may include various components, for example, a catheter assembly which includes a collapsed valve prosthesis, a guide wire and a plurality of sheaths and tubes. In order to provide percutaneous access for the objects into the patient's vasculature, an introducer is typically employed. Conventional introducers can comprise a hub component, a hemostasis valve, and an elongated tubular member which forms a lumen that runs through a central axis of the elongated tubular member through which the objects enter and exit the patient's vasculature.

[0004] During an aortic valve replacement procedure, introducers can accommodate objects of various sizes and shapes while also providing and maintaining support for the objects when they are inserted, removed and/or adjusted. Furthermore, introducers can be reliably percutaneously inserted without damaging the patient's vasculature while reducing the complexity of introducing and removing objects from the vasculature during surgical procedures.

[0005] In light of the above, a need exists for an expandable introducer with a flexible tubular member having the capability of expanding and contracting to accommodate objects during medical procedures while maintaining an atraumatic structure to minimize damage to the patient's vasculature.

SUMMARY

[0006] The following presents a simplified summary of the disclosure to provide a basic understanding of some aspects described in the detailed description.

[0007] Features of the present disclosure provide an expandable introducer apparatus having a flexible tubular member that seals to an outer surface of an object as it passes therethrough. Providing an introducer apparatus with the flexible tubular member can allow a clinician to reliably insert and remove objects from the vasculature of a patient while reducing injury to the patient's vasculature and reducing or eliminating blood loss.

[0008] In aspects, an expandable introducer apparatus comprises an inner tubular layer comprising an inner lumen and a radially expandable region extending along an elongated axis of the inner lumen and configured to at least partially radially expand relative to the elongated axis from a contracted orientation to an expanded orientation. The expandable introducer apparatus further comprises an outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer. The expandable introducer apparatus still further comprises a flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is

bonded to an outer surface of the distal segment of the inner tubular layer. The flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment. Furthermore, the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member. [0009] In further aspects, a method of guiding an object through an inner lumen of an expandable introducer apparatus is provided. The expandable introducer apparatus comprises an inner tubular layer comprising the inner lumen and a radially expandable region extending along an elongated axis of the inner lumen. The expandable introducer apparatus further comprises an outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer. The expandable introducer apparatus further comprises a flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer. The flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment. Furthermore, the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member. The method comprises distally advancing the object through a proximal end of the inner lumen, wherein a portion of the radially expandable region radially expands from a contracted orientation to an expanded orientation while a corresponding portion of the outer tubular layer expands to accommodate the expanded orientation of the radially expandable region. The method further comprises distally advancing the object past the distal end of the outer tubular layer to radially expand the radially expandable region of the inner tubular layer along the axial length of the distal segment while simultaneously circumferentially stretching the flexible region of the flexible tubular member, wherein the distal segment of the inner tubular layer and the flexible tubular member simultaneously expand to accommodate the object passing through the distal segment of the inner tubular layer. The method further comprises distally advancing the object past a distal end of the inner tubular layer.

[0010] Additional features and advantages of the aspects disclosed herein will be set forth in the detailed description that follows, and in part will be clear to those skilled in the art from that description or recognized by practicing the aspects described herein, including the detailed description which follows, the claims, as well as the appended drawings. It is to be understood that both the foregoing general description and the following detailed description present aspects intended to provide an overview or framework for understanding the nature and character of the aspects disclosed herein. The accompanying drawings are included to provide further understanding and are incorporated into and constitute a part of this specification. The drawings illustrate various aspects of the disclosure, and together with the description explain the principles and operations thereof.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] These and other features, aspects and advantages are better understood when the following detailed description is read with reference to the accompanying drawings, in which:

[0012] FIG. 1 is a schematic view of an exemplary expandable introducer apparatus in accordance with aspects of the present disclosure;

[0013] FIG. 2 is a schematic enlarged view of a distal end portion of the expandable introducer apparatus taken at view 2 of FIG. 1 in accordance with aspects of the present disclosure;

[0014] FIG. 3 is top schematic view of the distal end portion of the expandable introducer apparatus along line 3-3 of FIG. 2;

[0015] FIG. 4 is a schematic cross-sectional view of the distal end portion taken at line 4-4 of FIG. 2;

[0016] FIG. 5 is a schematic cross-sectional view of a distal segment of the distal end portion taken at line 5-5 of FIG. 2;

[0017] FIG. 6 is a schematic cross-sectional view of another embodiment of a distal segment of the distal end portion taken at line 5-5 of FIG. 2;

[0018] FIG. 7 is a schematic cross-sectional view of another embodiment of a distal segment of the distal end portion taken at line 5-5 of FIG. 2;

[0019] FIG. 8 is a schematic cross-sectional view of a flexible tip taken at line 8-8 of FIG. 2;

[0020] FIGS. 9-12 illustrate example steps in methods of manufacturing the expandable introducer apparatus in accordance with the present disclosure;

[0021] FIG. 13 schematically illustrates an object passing through an inner lumen of an inner tubular layer wherein a portion of a radially expandable region of the inner tubular layer radially expands;

[0022] FIG. 14 is a cross-sectional view taken at line 14-14 of FIG. 13, wherein an elongated split of the outer tubular layer is shown to separate to accommodate the expanded orientation of the radially expandable region;

[0023] FIG. 15 schematically illustrates the object passing through a distal segment of the inner tubular layer, wherein the distal segment of the inner tubular layer and a flexible tubular member simultaneously expand to accommodate the object passing through the distal segment;

[0024] FIG. 16 is a cross-sectional view taken at line 15-15 of FIG. 15;

[0025] FIG. 17 schematically illustrates the expandable introducer apparatus in accordance with the present disclosure after the object has passed through a distal end of the expandable introducer; and

[0026] FIG. 18 illustrates an exemplary flow chart demonstrating a method of utilizing the expandable introducer apparatus.

DETAILED DESCRIPTION

[0027] Aspects will now be described more fully hereinafter with reference to the accompanying drawings in which example aspects are shown. Whenever possible, the same reference numerals are used throughout the drawings to refer to the same or like parts. However, this disclosure may be embodied in many different forms and should not be construed as limited to the aspects set forth herein.

[0028] As used herein, the term “about” means that amounts, sizes, formulations, parameters, and other quantities and characteristics are not, and need not be, exact, but may be approximate and/or larger or smaller, as desired, reflecting tolerances, conversion factors, rounding off, measurement error and the like, and other factors known to those of skill in the art.

[0029] Ranges can be expressed herein as from “about” one value, and/or to “about” another value. When such a range is expressed, aspects include from the one value to the other value. Similarly, when values are expressed as approximations by use of the antecedent “about,” it will be understood that the value forms another aspect. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint.

[0030] Directional terms as used herein—for example up, down, right, left, front, back, top, bottom, upper, lower, etc.—are made only with reference to the figures as drawn and are not intended to imply absolute orientation.

[0031] Unless otherwise expressly stated, it is in no way intended that any methods set forth herein be construed as requiring that its steps be performed in a specific order, nor that with any apparatus, specific orientations be required. Accordingly, where a method claim does not actually recite an order to be followed by its steps, or that any apparatus claim does not actually recite an order or orientation to individual components, or it is not otherwise specifically stated in the claims or description that the steps are to be limited to a specific order, or that a specific order or orientation

to components of an apparatus is not recited, it is in no way intended that an order or orientation be inferred in any respect. This holds for any possible non-express basis for interpretation, including matters of logic relative to arrangement of steps, operational flow, order of components, or orientation of components; plain meaning derived from grammatical organization or punctuation, and; the number or type of aspects described in the specification.

[0032] As used herein, the singular forms “a,” “an,” and “the” include plural references unless the context clearly dictates otherwise. Thus, for example, reference to “a” component includes aspects having two or more such components, unless the context clearly indicates otherwise.

[0033] The word “exemplary,” “example,” or various forms thereof are used herein to mean serving as an example, instance, or illustration. Any aspect or design described herein as “exemplary” or as an “example” should not be construed as preferred or advantageous over other aspects or designs. Furthermore, examples are provided solely for purposes of clarity and understanding and are not meant to limit or restrict the disclosed subject matter or relevant portions of this disclosure in any manner. It can be appreciated that a myriad of additional or alternate examples of varying scope could have been presented but have been omitted for purposes of brevity.

[0034] As used herein, the terms “comprising,” “including,” and variations thereof shall be construed as synonymous and open-ended, unless otherwise indicated. A list of elements following the transitional phrases comprising or including is a non-exclusive list, such that elements in addition to those specifically recited in the list may also be present.

[0035] The terms “substantial,” “substantially,” and variations thereof as used herein are intended to represent that a described feature is equal or approximately equal to a value or description. Moreover, “substantially” is intended to denote that two values are equal or approximately equal. The term “substantially” may denote values within about 10% of each other, for example, within about 5% of each other, or within about 2% of each other.

[0036] Modifications may be made to the instant disclosure without departing from the scope or spirit of the claimed subject matter. Unless specified otherwise, “first,” “second,” or the like are not intended to imply a temporal aspect, a spatial aspect, an ordering, etc. Rather, such terms are merely used as identifiers, names, etc. for features, elements, items, etc. For example, a first end and a second end generally correspond to end A and end B or two different ends.

[0037] Unless otherwise indicated, the terms “distal” and “proximal” are used in the following description with respect to a position or direction relative to the treating clinician. “Distal” and “distally” are positions distant from or in a direction away from the clinician, and “proximal” and “proximally” are positions near or in a direction toward the clinician. In addition, the term “self-expanding” may be used in the following description with reference to one or more valve or stent structures of the prostheses hereof and is intended to convey that the structures are shaped or formed from a material that can be provided with a mechanical memory to return the structure from a compressed or constricted delivery configuration to an expanded deployed configuration or vice versa. Non-exhaustive exemplary self-expanding materials include stainless steel, a pseudo-elastic metal such as a nickel titanium alloy or nitinol, various polymers, or a so-called super alloy, which may have a base metal of nickel, cobalt, chromium, or other metal. Mechanical memory may be imparted to a wire or stent structure by thermal treatment to achieve a spring temper in stainless steel, for example, or to set a shape memory in a susceptible metal alloy, such as nitinol. Various polymers that can be made to have shape memory characteristics may also be suitable for use in aspects hereof to include polymers such as polynorborene, trans-polyisoprene, styrene-butadiene, and polyurethane. As well poly L-D lactic copolymer, oligo caprylactone copolymer and poly cyclo-octine can be used separately or in conjunction with other shape memory polymers.

[0038] Diseases associated with heart valves, such as those caused by damage or a defect, can include stenosis and valvular insufficiency or regurgitation. For example, valvular stenosis causes the valve to become narrowed and hardened which can prevent blood flow to a downstream heart

chamber from occurring at the proper flow rate and may cause the heart to work harder to pump the blood through the diseased valve. Valvular insufficiency or regurgitation occurs when the valve does not close completely, allowing blood to flow backwards, thereby causing the heart to be less efficient. A diseased or damaged valve, which can be congenital, age-related, drug-induced, or in some instances, caused by infection, can result in an enlarged, thickened heart that loses elasticity and efficiency. Some symptoms of heart valve diseases can include weakness, shortness of breath, dizziness, fainting, palpitations, anemia and edema, and blood clots which can increase the likelihood of stroke or pulmonary embolism. Symptoms can often be severe enough to be debilitating and/or life threatening.

[0039] Heart valve prostheses have been developed for repair and replacement of diseased and/or damaged heart valves. Such heart valve prostheses can be percutaneously delivered and deployed at the site of the diseased heart valve through catheter-based delivery systems. Such heart valve prostheses generally include a frame or stent and a prosthetic valve mounted within the frame. Such heart valve prostheses are delivered in a radially compressed or crimped configuration so that the heart valve prosthesis can be advanced through the patient's vasculature. Once positioned at the treatment site, the heart valve prosthesis is expanded to engage tissue at the diseased heart valve region to, for instance, hold the heart valve prosthesis in position.

[0040] In order to facilitate the introduction of the catheter-based delivery system including the heart valve prostheses and various other objects utilized during a procedure, an expandable introducer is first inserted through the skin of the patient. The expandable introducer is typically one which has an expandable region to accommodate the passage of the objects therethrough.

[0041] FIG. 1 illustrates an expandable introducer **101** for use with a plurality of objects, such as for example, a catheter-based delivery system. The expandable introducer **101** can be delivered to a desired location within a patient's vasculature for inserting objects therethrough. As shown in FIG. 1, the expandable introducer **101** can comprise an outer tubular layer **401**, an inlet **105**, and a flush port **111**. The object can be inserted through the proximal end of the inlet **105** and then advanced distally along a length of the expandable introducer. The object then exits the expandable introducer **101** through a distal end portion of the expandable introducer **101** where the object is thereafter advanced within the vasculature to the treatment site.

[0042] A distal end portion **103** of the expandable introducer **101** is illustrated and discussed with initial reference to FIGS. 2-5. As shown in FIGS. 4-5, an inner tubular layer **403** comprises an inner lumen **405** and a radially expandable region **407** (see FIG. 4) extending along an elongated axis **201** of the inner lumen **405**. The radially expandable region **407** may be configured to at least partially radially expand relative to the elongated axis **201** from a contracted orientation (e.g., see FIGS. 4-5) to an expanded orientation (e.g., see FIGS. 14 and 16). The inner tubular layer **403** may be a wide range of materials. For example, the inner tubular layer **403** may be formed from a flexible material, such as for example, tetrafluoroethylene (TFE), Teflon®, polytetrafluoroethylene (PTFE), polyethylene, polyethylene terephthalate (PET), polyester or the like. Further, the inner tubular layer **403** may have a low coefficient of friction to aid in the insertion of objects therethrough.

[0043] In some aspects, the radially expandable region **407** of the inner tubular layer **403** can comprise a foldable region **409** that may be configured to at least partially unfold from a contracted orientation to an expanded orientation. In some aspects, the foldable region **409** further comprises an outer fold **411** and an inner fold **413** in the contracted orientation as shown in FIG. 4. The outer fold **411** and the inner fold **413** are configured such that when an object **1701** (e.g., FIG. 14) passes through the inner lumen **405**, the outer fold **411** and the inner fold **413** begin to unfold. For example, the outer fold **411** and the inner fold **413** can partially or completely unfold to allow the inner tubular layer **403** to radially expand to accommodate the object **1701** passing through the inner lumen **405**.

[0044] Throughout the disclosure, the various embodiments of the expandable introducer is

illustrated and described being used to introduce the object **1701** schematically illustrated in the drawings. The schematic illustration of the object **1701** is not intended to be limited in size and shape to the illustrated size and shape but may take the size and shape of the object being introduced. Such objects may comprise various types of prosthetic heart valves although the object may comprise other prostheses such as a prosthetic bone fragment, or other prosthetic implant. In further examples, the object may comprise pacemakers or other medical devices or medical implants or other surgical objects to be percutaneously delivered into inserted into the vasculature of the patient.

[0045] In some aspects, the expandable introducer **101** can further comprise an outer tubular layer **401** comprising an inner surface **417** bonded to the inner tubular layer **403**. In some aspects, as shown in FIGS. **3-4**, the outer tubular layer **401** can comprise an optional elongated split **301** extending along the radially expandable region **407** in a direction of the elongated axis **201**. As shown in FIG. **4**, the optional elongated split **301** can be formed by a first edge facing a second edge. As shown, the first edge can comprise an unobstructed view of the second edge wherein an empty space exists between the first edge and the second edge when the radially expandable region **407** of the inner tubular layer **403** is in the contracted orientation. Although not shown, in some embodiments, the first edge and the second edge of the optional elongated split **301** may abut one another or overlap one another in the contracted orientation in further embodiments. Furthermore, as shown, the optional elongated split **301** can extend along the radially expandable region **407** in the direction of the elongated axis **201** while the radially expandable region **407** of the inner tubular layer **403** is positioned entirely radially inward relative to the optional elongated split **301**. Although not shown, in further embodiments, the radially expandable region **407** of the inner tubular layer **403** may partially extend within the optional elongated split **301** wherein the radially expandable region **407** partially or entirely obstructs the view of the first edge from the second edge. Still further, the first edge and the second edge are illustrated as parallel with respect to one another although the first edge (or portions of the first edge) may not be parallel to the second edge (or portions of the second edge) in further embodiments. As further illustrated, the first edge and the second edge are each linear edges that are parallel with one another and each extend in the direction of the elongated axis **201**. In further embodiments, although not shown, the first edge and the second edge may comprise curvilinear or other non-linear edge shapes.

[0046] The optional elongated split **301** can be configured such that when an object **1701** passes through the inner lumen **405**, the optional elongated split **301** circumferentially expands to allow the object **1701** to pass therethrough. As shown by joining segment **402** schematically illustrated in FIG. **4**, any of the embodiments of the disclosure may provide the outer tubular layer **401** without the previously-described elongated split. Rather, as schematically represented by the joining segment **402**, the outer tubular layer **401** may comprise a cross section taken perpendicular to the elongated axis **201** that is continuous around the entire periphery without any breaks that would otherwise be caused by an elongated split. Furthermore, such continuity can extend along a substantial portion of the length, such as the entire length, of the outer tubular layer **401** in a direction of the elongated axis **201**. Furthermore, while the joining segment **402** is illustrated as having a thickness that is substantially the same as the thickness as the remainder of the periphery of the outer tubular layer, different thicknesses may be provided in further embodiments. For instance, the thickness of the joining segment **402** can be thinner at the radially expandable region **407** to allow easier expansion of the joining segment **402** at the foldable region **409** during expansion of the inner tubular layer **403**.

[0047] The outer tubular layer **401** may be a wide range of materials. For example, the outer tubular layer **401** may be formed from a shape memory material or a resilient material, such as for example, polyurethane (e.g., Pellethane®, Elasthane™, Texin®, or Tecothane®), Pebax®, polyethylene or the like. Additionally, and/or alternatively, the outer tubular layer **401** may further include material(s) that allow the outer tubular layer **401** to be radiopaque to facilitate the detection

of the expandable introducer **101** by electromagnetic radiation.

[0048] Referring to FIGS. **3** and **5**, in some aspects, a distal segment **1001** of the inner tubular layer **403** extends beyond a distal end **306** of the outer tubular layer **401**. In some aspects, the expandable introducer **101** further comprises a flexible tubular member **300** that comprises a proximal end **305** that can abut, and in some examples engage, the distal end **306** of the outer tubular layer **401** although the proximal end **305** of the flexible tubular member **300** may not abut the distal end **306** of the outer tubular layer **401** (e.g., may be spaced apart) in further embodiments. The flexible tubular member **300** further comprises a distal end **307**, and a length **L1** extending from the proximal end **305** to the distal end **307** in a direction of the elongated axis **201**. For example, the length **L1** can be from about 5 mm to about 7 mm. In aspects, the flexible tubular member **300** can comprise a flexible tip **313** that distally extends beyond a distal end **315** of the inner tubular layer **403**. The flexible tip **313** can comprise a length **L2** extending from the distal end **315** of the inner tubular layer **403** to the distal end **307** of the flexible tubular member **300** in the direction of the elongated axis **201**. For example, the length **L2** of the flexible tip **313** can be from about 0.5 mm to about 3 mm, for example, from about 1 mm to about 1.5 mm.

[0049] The flexible tubular member **300** may be made from a variety of suitable materials. For example, the flexible tubular member **300** can be made from a highly flexible elastic material having a high columnar strength. High columnar strength refers to the ability of the material to withstand buckling or collapsing. In this way, the material should permit the flexible tubular member **300** and the flexible tip **313** to expand and contract, enabling them to accommodate the object **1701** passing therethrough while being capable of preventing the flexible tip **313** from collapsing. For example, a high columnar strength will prevent the flexible tip **313** from collapsing inward towards the proximal end **305** of the flexible tubular member **300** when the object **1701** is proximally retracted relative to the flexible tubular member **300** when removing the object **1701** from the patient's vasculature and from the expandable introducer **101**. The material can be a wide range of materials. For example, the flexible tubular member **300** may be made from Thermoplastic Polyurethane (TPU), Pellethane, Texin, and/or any other similarly suitable materials.

[0050] As shown in FIG. **5**, in some aspects the flexible tubular member **300** comprises a bonded region **501** and a flexible region **311** (schematically represented by dashed lines). The bonded region **501** and flexible region **311** are each schematically represented by dashed lines in FIG. **5** that are intended to represent an example of the circumferential extent of the bonded and flexible regions at an arcuate interface between the flexible tubular member **300** and the distal segment **1001** of the inner tubular layer **403**. For example, bonded region **501** is understood to comprise an inner surface **503** of the flexible tubular member **300** bonded to an outer surface **504** of the distal segment **1001** of the inner tubular layer **403** along the circumferential portion of the arcuate interface between the inner surface **503** and the outer surface **504** that is parallel to the illustrated arcuate broken line indicating the bonded region **501** of the arcuate interface. Furthermore, in some embodiments, the bonded region can extend a length (e.g., the entire length) of the distal segment **1001** of the inner tubular layer **403** in the direction of the elongated axis **201**.

[0051] As further shown in FIG. **5**, flexible region **311** is understood to comprise the inner surface **503** of the flexible tubular member **300** that is not bonded to the outer surface **504** of the distal segment **1001** of the inner tubular layer **403** along the circumferential portion of the arcuate interface between the inner surface **503** and the outer surface **504** that is parallel to the illustrated arcuate broken line indicating the flexible region **311** of the arcuate interface. As such, within the flexible region, the inner surface **503** of the flexible tubular member **300** is not bonded to the radially expandable region **407** of the inner tubular layer **403**. Furthermore, as shown in FIG. **3**, the flexible region **311**, in some embodiments, can extend a length (e.g., the entire length) of the distal segment **1001** of the inner tubular layer **403** in the direction of the elongated axis **201**.

[0052] In aspects, the radially expandable region **407** of the distal segment **1001** of the inner tubular layer **403** extends under the flexible region **311** of the flexible tubular member **300**. For

example, the radially expandable region **407** can extend under the flexible region **311** of the flexible tubular member **300** along the entire length of the distal segment **1001** in the direction of the elongated axis **201**. In this way, when the object **1701** passes through the inner lumen **405**, the flexible region **311** and the radially expandable region **407** will be able to simultaneously radially expand together to accommodate the object **1701** passing therethrough.

[0053] In the embodiment illustrated in FIG. 5, the flexible tubular member **300** surrounds the inner tubular layer **403** with an inner cross-sectional surface profile (see inner surface **503**) matching an outer cross-sectional surface profile (see outer surface **504**) of the distal segment **1001** of the inner tubular layer **403**. In this way, the flexible tubular member **300** can be bonded to the inner tubular layer **403** within the bonded region **501** as discussed above. In some aspects, the flexible tubular member **300** comprises a circular cross-sectional area **505** with a uniform wall thickness **507** circumscribing the distal segment **1001** of the inner tubular layer **403**.

[0054] In some aspects, the bonded region **501** can be formed utilizing a number of suitable bonding processes. For example, the bonding process may include but is not limited to a reflow bonding process or a laser bonding process, such as for example, by pressing the flexible tubular member **300** to the outer cross-sectional surface profile of the distal segment **1001** of the inner tubular layer **403** and applying a heat source to form the bonded region **501**. In this way, the bonded region **501** is formed where the flexible tubular member **300** becomes joined to the outer cross-sectional surface profile of the distal segment **1001** where the heat source is applied. The bonding procedure is not limited to a reflow or laser bonding process. For example, the bonding procedure may comprise an adhesive bonding process or a process that involves manufacturing the flexible tubular member **300** as a monolithic structure of the expandable introducer **101**.

[0055] FIG. 6 illustrates additional embodiments, of a flexible tubular member **600** comprising an inner circumferential flared segment **601** comprising a flared end **603** abutting a folded end of the outer fold **411** of the foldable region **409**. In some embodiments, the inner circumferential flared segment **601** may extend (e.g., continuously extend) the entire length “L1” of the flexible tubular member **600** (see “L1” in FIG. 3). Alternatively, the inner circumferential flared segment **601** may extend less than the entire length of the flexible tubular member **600**. For example, the inner circumferential flared segment **601** may extend along the distal segment **1001** of the inner tubular layer **403** without extending into the length L2 of the flexible tip **313**.

[0056] As further shown in FIG. 6, the flared end **603** can comprise a thickness that gradually circumferentially increases (e.g., in the illustrated counterclockwise direction) toward the outer fold **411**. Further, the outer fold **411** circumferentially extends toward the flared end **603** of the flared segment (e.g., in the illustrated clockwise direction) such that the folded end of the outer fold **411** faces the flared end **603**. As shown, in some embodiments, the folded end of the outer fold **411** can abut the flared end **603** wherein the flared end **603** can provide a seat for the folded end of the outer fold **411**. The inner circumferential flared segment **601** can fill a negative space that may otherwise be formed by the outer fold **411**. In this way, as shown in FIG. 6, the inner circumferential flared segment **601** may provide the outer circumferential surface of the flexible tubular member **600** with a more uniform profile and can support the flexible tubular member **600** to resist radial deformation and stress concentrations that may otherwise impact the profile of the flexible tubular member **600**. The circumferential flared segment **601** can therefore provide a more uniform or circumferential expansion of the flexible tubular member **600**.

[0057] FIG. 7 illustrates an additional embodiment of a flexible tubular member **700** comprising a first inner protrusion **701a** aligned with the folded end of the outer fold **411** of the foldable region **409** and a second inner protrusion **701b** aligned with the folded end of the inner fold **413** of the foldable region **409**. In some embodiments, the first and second inner protrusion **701a**, **701b** may extend (e.g., continuously extend) the entire length “L1” of the flexible tubular member **700** (see “L1” in FIG. 3). Alternatively, the first and second inner protrusion **701a**, **701b** may extend less than the entire length of the flexible tubular member **700**. For example, the first and second inner

protrusion **701a**, **701b** may extend along the distal segment **1001** of the inner tubular layer **403** without extending into the length **L2** of the flexible tip **313**.

[0058] As further illustrated in FIG. 7, the first and second inner protrusions **701a**, **701b** can comprise thickened areas of the flexible tubular member **700** that are spaced apart from one another by an intermediate area of reduced thickness relative to the first and second inner protrusions **701a**, **701b**. Furthermore, as shown, in some embodiments, a majority of the circumference of the flexible tubular member **700** can extend with a reduced thickness relative to the first and second inner protrusions **701a**, **701b** that circumferentially extends from the first inner protrusion **701a** to the second inner protrusion **701b**. As shown, the majority of the circumference of the flexible tubular member **700** and the intermediate area extending between the first and second inner protrusions **701a**, **701b** can comprise substantially the same reduced thickness although different thicknesses may be provided in further embodiments. As shown, the first inner protrusion **701a** can be aligned to press against the folded end of the outer fold **411** while the second inner protrusion **701b** can be aligned to press against the folded end of the inner fold **413**. In this way, a more uniform circumferential expansion of the flexible tubular member **700** can be maintained subsequent to the bonding process.

[0059] The flexible tubular member **300**, **600**, **700** can be manufactured utilizing any number of manufacturing methods. For example, the flexible tubular member **300**, **600**, **700** can be created utilizing an extrusion process, and/or a molding process. In some embodiments, the flexible tubular member **300**, **600**, **700** can be formed by radio-frequency molding, or radio-frequency forming of an extrusion. In this way, the flexible tubular member **300**, **600**, **700** can be manufactured to conform to a variety of material characteristics and design specifications, such as for example, a material with a high columnar strength and low modulus of elasticity to facilitate the expansion and contraction of the flexible region **311** while avoiding failure (e.g., collapsing).

[0060] As shown in the cross-section of FIG. 8, in some embodiments, the flexible tip **313** can comprise a circular cross-sectional area **800** with a uniform wall thickness **803** although nonuniform wall thicknesses may be provided in further embodiments. In other aspects, the flexible tip **313** may comprise any other shape(s) required to accommodate the object **1701** passing through the distal end **315** of the inner tubular layer **403**.

[0061] FIGS. 9-12 illustrate exemplary methods of manufacturing the expandable introducer **101** in accordance with the present disclosure. FIG. 9 illustrates a perspective view of the inner tubular layer **403** comprising the foldable region **409** comprising the outer fold **411** and the inner fold **413**. Further depicted is the inner lumen **405** extending along the elongated axis **201**. In aspects, the foldable region **409** can extend along the elongated axis **201** of the inner lumen **405** (e.g., along the entire length of the inner tubular layer **403** in the direction of the elongated axis **201**). In some aspects, a gap can be formed between the outer fold **411** and the inner fold **413** and extends along the elongated axis **201**.

[0062] As explained previously, during a reflow or laser bonding process, heat may be applied to surfaces of materials to be bonded. In this way, the layers of materials begin to melt as a result of the directed heat source. Accordingly, the surfaces of the materials begin to contact one another. Once the layers of the melted materials solidify, the layers form a strong bond with one another.

[0063] In aspects, in order to prevent the bonding of the foldable region **409**, for example, portions of the flexible tubular member **300** from melting and flowing into the inner fold **413** and/or portions of the inner fold **413** and the outer fold **411** melting together, an inner shim **910** can be placed within the gap between the outer fold **411** and the inner fold **413**. For example, the inner shim **910** can be slidably positioned through the gap between the outer fold **411** and the inner fold **413** and extend into the inner fold **413** from outside of the inner tubular layer **403**. The inner shim **910** can extend along the elongated axis **201** of the inner lumen **405** any length required to prevent a bonding of the inner fold **413** together and/or a bonding of the outer fold **411** to the inner fold **413**. For example, the inner shim **910** can extend a total length of the inner tubular layer **403**. In an

alternative embodiment, the inner shim **910** can extend half the length of the inner tubular layer **403**. In some examples, as shown in FIG. **10**, the inner shim **910** can have a length that is greater than the length of the distal segment **1001** in the direction of the elongated axis **201**. For example, as shown, in some embodiments, the proximal end of the inner shim **910** can extend proximally past the distal end **306** of the outer tubular layer **401** while also extending distally past the distal end **315** of the inner tubular layer **403**.

[0064] FIG. **10** illustrates a perspective view of the expandable introducer **101** comprising the outer tubular layer **401** positioned over the inner tubular layer **403** with the optional elongated split **301** of the outer tubular layer **401** aligned with the foldable region **409** of the inner tubular layer **403**. Although not illustrated in FIG. **10**, in further embodiments, the outer tubular layer **401** may comprise a circular cylindrical tube without an elongated split. Indeed, the outer tubular layer **401** may continuously extend as schematically represented by the joining segment **402** illustrated in FIGS. **4** and **13-14**. Furthermore, as discussed above, the joining segment **402** may have a reduced thickness and can be aligned with the foldable region **409** of the inner tubular layer **403** to facilitate expansion of the joining segment **402** at the location where the inner tubular layer **403** unfolds during expansion of the inner tubular layer **403**. As further illustrated, the outer tubular layer **401** can be positioned relative to the inner tubular layer such that the distal segment **1001** of the inner tubular layer **403** extends beyond the distal end **306** of the outer tubular layer **401**.

[0065] In some aspects, the flexible tubular member **300**, **600**, **700**, shown in FIGS. **2-3**, and **5-8**, can be bonded to the distal segment **1001**. The distal segment **1001** can be any number of suitable lengths. For example, the distal segment **1001** can be a length of 6 mm. In this way, the flexible tubular member **300**, **600**, **700** will have a 6 mm bonding length. In this non-limiting example, the proximal end **305** of the flexible tubular member **300** can abut the distal end **306** of the outer tubular layer **401** and extend distally in the distal direction along the elongated axis **201** of the inner lumen **405** as described in FIGS. **2-3**, and **5**.

[0066] FIG. **11** illustrates a perspective side view of the expandable introducer **101** with an outer shim **1102** comprising a proximal portion positioned on a distal portion of an outer surface **1103** of the outer tubular layer **401** and extending along the elongated axis **201** of the inner lumen **405** and over an outer area of the distal segment **1001**. For example, as shown, in some embodiments, the outer shim **1102** can extend over the outer area along the entire length of the distal segment **1001**. Due to the positioning of the outer shim **1102**, the flexible tubular member **300** will not bond to the distal segment **1001** of the inner tubular layer **403** at the outer area covered by the outer shim **1102** and the flexible tubular member will not bond the facing split edges of the outer tubular layer **401** together. In this way, the flexible region **311** will be created to allow the foldable region **409** to freely unfold and facing edges to spread apart so that the inner tubular layer **403** can achieve the expanded orientation. In some aspects, the bonded region **501** may be calculated to bond to half the circumference of the distal segment **1001** to the flexible tubular member **300**. For example, the circumferential extent of the bonded region **501** can be calculated by determining the circumference of the distal segment **1001** and dividing the determined circumference number in half. In other aspects, the outer shim **1102** can be positioned to cover the folded end of the outer fold **411** and the folded end of the inner fold **413**. In other aspects, the outer shim **1102** can be any other suitable width required to form the flexible region **311**.

[0067] In some aspects, the outer shim **1102** can be a variety of suitable lengths. For example, in aspects, the outer shim **1102** can correspond to the length of the distal segment **1001**, such as for example, in some embodiments the distal segment **1001** can be 6 mm from the distal end **306** of the outer tubular layer **401** and the outer shim **1102** can also be 6 mm and extend distally from the distal end **306** of the outer tubular layer **401**. In other non-limiting examples, the outer shim **1102** can be any other length necessary to form the flexible region **311**. In the illustrated embodiment, the outer shim **1102** can be longer than the length of the distal segment **1001** to facilitate removal of the outer shim **1102** after bonding the flexible tubular member to the distal segment **1001**. For

example, as mentioned previously, a proximal portion of the outer shim **1102** can be positioned on the distal portion of the outer surface **1103** of the outer tubular layer **401**. At the same time, in some embodiments, the outer shim can extend along the elongated axis **201** of the inner lumen **405** while extending over the outer area of the distal segment, and distally past the distal end **315** of the inner tubular layer **403**. The outer shim **1102** can therefore prevent undesired bonding at the foldable region **409** and the distal portion of the elongated split **301**, if provided. As such, once formed, the foldable region **409** is free to unfold while the facing edges of the elongated split **301**, if provided, are free to spread apart so that the inner tubular layer **403** can achieve the expanded orientation.

[0068] FIG. **12** illustrates a perspective view of the expandable introducer **101** shown in FIG. **11** with the flexible tubular member **300**, **600**, **700** bonded to the distal segment **1001** of the inner tubular layer **403**. As shown, the flexible tip **313** of the flexible tubular member **300**, **600**, **700** extends beyond the distal end **315** of the inner tubular layer **403**. The inner shim **910** and outer shim **1102** have been removed since the bonding process has been completed. For example, after bonding the flexible tubular member **300**, **600**, **700** to the distal segment **1001**, the inner shim **910** may be gently pulled out from between the gap of the outer fold **411** and the inner fold **413**, utilizing extra precaution to avoid tearing or ripping the outer fold **411** or the inner fold **413**. The inner shim **910** and the outer shim **1102** may be any number of suitable materials that prevent the previously-described bonding. For example, the inner shim **910** and/or the outer shim **1102** can be made from a thin sheet of ceramic, metal, polyimide, PTFE and/or the like.

[0069] FIGS. **13-18** will now describe a method of guiding an object **1701** through the expandable introducer **101** comprising the flexible tubular member **300**, **600**, **700** with the understanding that similar or identical methods may be provided in the other embodiments of the disclosure. More specifically, FIG. **18** illustrates an exemplary flow chart showing example steps for guiding the object **1701** through the expandable introducer **101**.

[0070] Referring to FIG. **18**, during step **1801**, an incision is made at an anatomical location of the patient's vessel. For example, a transfemoral approach that involves making an incision in the groin and operates as a passageway for guiding a catheter-based delivery system to a target site (e.g., a diseased or defective heart valve).

[0071] As shown during step **1803**, the method can further comprise percutaneously introducing a guide wire into the patient's vasculature via the incision made during step **1801**. The guide wire can then be carefully guided through the patient's vasculature to the target site. For example, the guide wire can be guided through the incision in the groin of the patient to a the location of the diseased or defective heart valve (e.g., aortic valve). The guide wire can serve as a path for subsequent objects to travel along when inserted into the patient via the incision made during step **1801**.

[0072] During step **1805**, the method can comprise percutaneously inserting the expandable introducer **101** into the incision made during step **1801** and sliding the expandable introducer **101** into the patient's vasculature. For example, the expandable introducer **101** can slide along the guide wire introduced during step **1803**. In some aspects, the expandable introducer **101** can have a dilator introduced within the inner lumen **405** to facilitate the expansion of a passageway within the patient's vessel. As shown, in some embodiments, the flexible tip **313** can comprise a constricted conical shape. In some aspects, the constricted conical shape of the flexible tip **313** axially covers the distal end **315** of the inner tubular layer **403** and thereby provides a beneficial atraumatic feature that helps avoid injury or irritation to the patient's vasculature when inserting the expandable introducer **101**.

[0073] Next, during step **1807** the method can comprise distally advancing the object **1701** through the proximal end of the inlet **105** into the proximal end **309** of the inner lumen **405**. As shown in FIGS. **13-14**, a portion **1401** of the radially expandable region **407** radially expands from a contracted orientation to an expanded orientation while a corresponding portion of the outer tubular layer **401** expands to accommodate the expanded orientation or the radially expandable region of the inner tubular layer **403**. In embodiments with an elongated split **301**, the expansion of the

corresponding portion of the outer tubular layer **401** can comprise separating the elongated split **301**, as shown in FIGS. **13-14**, to accommodate the expanded orientation of the inner tubular layer **403**. The object **1701** is schematically represented in cross-section in FIGS. **14** and **16**. Alternatively, if the outer tubular layer **401** is not provided with an elongated split, a portion of the outer tubular layer **401** (e.g., the joining segment **402**) or the entire circumference of the outer tubular layer **401** may expand to accommodate the expanded orientation of the inner tubular layer **403**.

[0074] As shown in FIG. **14**, in embodiments with an elongated split **301**, the radially expandable region **407** of the inner tubular layer **403** can radially expand while the facing edges of the elongated split **301** of the outer tubular layer **401** circumferentially spread apart such that the inner tubular layer **403** achieves the expanded orientation to accommodate the object **1701**. Indeed, as shown, the facing edges of the elongated split **301** can spread apart from one another such that the width of the split increases to accommodate the expansion of the inner tubular layer **403**. The portion **1401** appears bulged as a result of the object **1701** passing through the inner lumen **405** and thus conforms to the shape of the object **1701** passing therethrough. Alternatively, if the outer tubular layer **401** is provided as a continuous tube (e.g., with the joining segment **402**) without the elongated split, portions of the outer tubular layer **401** or the entire outer circumference of the outer tubular layer **401** can stretch to accommodate the expansion of the inner tubular layer **403**. For example, the circumferential width of the joining segment **402** can increase from the width shown in FIG. **4** to the expanded width shown in FIG. **14** to accommodate expansion of the inner tubular layer **403**. In further embodiments, the entire circumferential width of the outer tubular layer **401** (e.g., not just a portion of the outer tubular layer in the vicinity of the foldable region **409**) can stretch to accommodate the expansion of the inner tubular layer **403**. As illustrated in FIG. **14**, the foldable region **409** has fully unfolded. For example, the outer fold **411** and the inner fold **413** have fully unfolded such that the inner lumen **405** has fully radially expanded to its maximum diameter. In some other aspects, the outer fold **411** and the inner fold **413** can partially radially expand (e.g., wherein the inner lumen **405** only partially radially expands). The foldable region **409** can expand to any range of suitable diameters. For example, the foldable region **409** can have an outer fold **411** and an inner fold **413** configured to accommodate the largest diameter object **1701** expected to pass through the inner lumen **405**. For example, a fold length between the outer fold **411** and the inner fold **413** can be increased (e.g., by forming a larger outer fold **411** and/or inner fold **413**). In this way, the maximum expanded outer diameter of the inner tubular layer **403** will be increased, thereby allowing larger objects to pass through the inner lumen **405**. As further illustrated, the elongated split **301** has separated (or the joining segment **402** has expanded) to accommodate the expansion of the radially expandable region.

[0075] After the object **1701** has been introduced into the expandable introducer **101**, during step **1809** the method can subsequently comprise distally advancing the object **1701** past the distal end **306** of the outer tubular layer **401**. The radially expandable region **407** of the inner tubular layer **403** then radially expands along the axial length of the distal segment **1001** while simultaneously circumferentially stretching the flexible region **311** of the flexible tubular member **300, 600, 700**. As such, once the object **1701** is advanced past the distal end **306**, the distal segment **1001** of the inner tubular layer **403** and the flexible tubular member **300, 600, 700** simultaneously expand to accommodate the object **1701** passing through the inner lumen **405** of the distal segment **1001** of the inner tubular layer **403**. For example, FIGS. **15-16** show the object **1701** being distally advanced past the distal end **306** of the outer tubular layer **401**, wherein a portion **1601** of the expandable introducer **101** radially expands as a result of the expansion of the radially expandable region **407** and the flexible region **311**.

[0076] During step **1811**, the method can further comprise distally advancing the object **1701** past the distal end **315** of the inner tubular layer **403**. For example, after the object **1701** has been distally advanced past the distal end **315** of the inner tubular layer **403**, the object **1701** will begin

entering the vasculature of the patient. In some examples, the object **1701** can move past the distal end **315** of the inner tubular layer **403** via a guide wire. In other examples, the object **1701** can move past the distal end **315** of the inner tubular layer **403** and into the vasculature of the patient without a guide wire.

[0077] Next, during step **1813**, as shown in FIG. **17**, subsequent to an enlarged end portion **1702** of the object **1701** passing through the distal end **315** and the inner tubular layer **403**, the flexible tubular member **300**, **600**, **700** and the inner tubular layer **403** contract towards a substantially similar original orientation as they were prior to the object **1701** passing through. For example, the resiliency of the material allows for the flexible tubular member **300**, **600**, **700** and the inner tubular layer **403** to recontract towards their original orientation. In this way, by allowing the flexible tubular member **300**, **600**, **700** and the inner tubular layer **403** to recontract towards their original orientation, the expandable introducer **101** can accommodate a variety of objects featuring a range of dimensions and structural designs (e.g., a catheter-based delivery system, a dilator and/or the like). For example, the recontraction allows for subsequent entry of a smaller object **1701** following the removal of a larger object **1701** (e.g., where the larger object **1701** has a greater diameter than the smaller object **1701**) while reducing or eliminating blood loss.

[0078] During step **1815**, as further illustrated by FIG. **17**, the method can further comprise distally advancing the object **1701** through the flexible tip **313** and past the distal end **307** of the flexible tubular member **300**, **600**, **700**. In some aspects, the flexible tip **313** forms a rounded transition **1703** at an interface with an outer circumferential surface **1707** of the object **1701** when advancing the object **1701** past the distal end **307** as well as forming a rounded transition at the distal end **315** of the inner tubular layer **403**. The rounded transitions **1703** can provide a more atraumatic feature compared to a transition with sharp edges, thereby avoiding irritation and possible vascular damage from the expandable introducer **101**. In some aspects, the distal end **307** of the flexible tubular member **300**, **600**, **700** forms a circumferential seal **1705** against the outer circumferential surface **1707** of the object **1701** as the object **1701** passes through the flexible tip **313** and past the distal end **307** of the flexible tubular member **300**, **600**, **700**. The circumferential seal **1705** in combination with the recontraction of the flexible tubular member **300**, **600**, **700** and the inner tubular layer **403** (as explained with reference to step **1813**) will be especially beneficial in preventing blood loss when an object **1701** is inserted, removed, repositioned, rotated and/or otherwise adjusted. In some aspects, the flexible tip **313** can be formed in the shape of a hollow frustum in order to form the above-referenced atraumatic transitions (e.g., the rounded transition **1703**, and the transition at the distal end **315** of the inner tubular layer **403**).

[0079] As shown in step **1817**, the method can further comprise positioning the object **1701** within a location of the patient. For example, in some aspects, the object **1701** can be a catheter delivery system, and the catheter delivery system can be inserted through the inner lumen **405** and positioned within a location of the patient's heart, for example, the location of the aortic valve (e.g., the position between the left ventricle and the aorta). Subsequently, the catheter-based delivery system can be deployed, such as for example, releasing a valve retention member from the catheter-based delivery system to deploy a prosthesis into the patient's heart.

[0080] In other embodiments, the method can further comprise proximally retracting the object **1701** within the inner lumen **405** proximally past the distal end **306** of the outer tubular layer **401** and out of the inner lumen **405** of the distal segment **1001** of the inner tubular layer **403**. Referring to FIG. **17**, for example, the enlarged end portion **1702** of the object **1701** can be proximally retracted relative to the flexible tubular member **300**, **600**, **700**. The flexible tip **313** can be designed to maintain the circumferential seal **1705** against the outer circumferential surface **1707** to inhibit (e.g., prevent) blood from entering into the inner lumen **405**. The flexible tip **313** can be designed to resist buckling and/or inversion when the enlarged distal end of the object **1701** reaches the distal end **307** of the flexible tubular member **300**, **600**, **700**. Indeed, the distal tip can be fabricated from a material that has a sufficiently high axial stiffness to avoid buckling and/or

inversion. Furthermore, the length “L2” can be kept sufficiently low to increase the axial stiffness as compared to longer lengths. For example, the length “L2” of the flexible tip **313** can be from about 0.5 mm to about 3 mm, for example, from about 1 mm to about 1.5 mm to avoid bucking and/or inversion when the enlarged distal end of the object **1701** is proximally retracted into the inner lumen **405** to cause the distal segment **1001** and the flexible tubular member **300, 600, 700** to radially expand to accommodate the object passing through the distal segment **1001** as shown in FIGS. 15-16.

[0081] Then, as shown in FIGS. 13-14, the object **1701** can be further proximally retracted past the distal end **306** of the outer tubular layer **401**. Due to the resiliency of the flexible tubular member **300, 600, 700**, the distal segment **1001** and the flexible tubular member **300, 600, 700** can then radially retract to the original nonexpanded state as shown in FIG. 13. As further shown in FIGS. 13-14, the foldable region **409** can unfold to radially expand the portion **1401** of the inner tubular layer **403** while the facing split edges of the elongated split **301** can circumferentially spread apart to further allow passage of the object **1701**. In embodiments without an elongated split **301**, the joining segment **402** or the entire circumference of the outer tubular layer **401** can stretch to further allow passage of the object **1701**. The object **1701** can be further proximally retracted until the object is removed from the proximal end of the expandable introducer **101**. Due to the resiliency of the outer tubular layer **401**, once the object **1701** has passed, the facing edges of the elongated split **301** can be drawn back together (e.g., to the position shown in FIG. 4), wherein the foldable region **409** regains the folded orientation to radially contract the inner tubular layer **403** as shown in FIG. 4. In embodiments without the elongated split, the outer tubular layer **401** can radially contract to draw back together, wherein the foldable region **409** regains the folded orientation to radially contract the inner tubular layer **403** shown in FIG. 4.

[0082] In accordance with the disclosure, non-limiting aspects of the disclosure will now be described. Various combinations of the aspects can be provided in accordance with the disclosure.

[0083] Aspect 1. An expandable introducer apparatus comprising an inner tubular layer comprising an inner lumen and a radially expandable region extending along an elongated axis of the inner lumen and configured to at least partially radially expand relative to the elongated axis from a contracted orientation to an expanded orientation. An outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer. A flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer, the flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment, and the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member. [0084] Aspect 2. The expandable introducer apparatus of Aspect 1, wherein the flexible tubular member comprises a proximal end, a distal end, and a length extending from the proximal end to the distal end in a direction of the elongated axis. [0085] Aspect 3. The expandable introducer apparatus of Aspect 2, wherein the length is from about 5 mm to about 7 mm. [0086] Aspect 4. The expandable introducer apparatus of any one of Aspects 2-3, wherein the proximal end of the flexible tubular member abuts the distal end of the outer tubular layer. [0087] Aspect 5. The expandable introducer apparatus of any one of Aspects 2-4, wherein the flexible tubular member comprises a flexible tip that extends beyond a distal end of the inner tubular layer. [0088] Aspect 6. The expandable introducer apparatus of Aspect 5, wherein the flexible tip comprises a length extending from the distal end of the inner tubular layer to the distal end of the flexible tubular member. [0089] Aspect 7. The expandable introducer apparatus of Aspect 6, wherein the length of the flexible tip is from about 0.5 mm to about 3 mm. [0090] Aspect 8. The expandable introducer apparatus of Aspect 7, wherein the length of the flexible tip is from about 1 mm to about 1.5 mm. [0091] Aspect 9. The expandable introducer apparatus of any one of Aspects 1-8, wherein the radially expandable region of the inner

tubular layer comprises a foldable region configured to be at least partially unfolded from the contracted orientation to the expanded orientation. [0092] Aspect 10. The expandable introducer apparatus of Aspect 9, wherein the foldable region further comprises an outer fold and an inner fold in the contracted orientation. [0093] Aspect 11. The expandable introducer apparatus of Aspect 10, wherein the flexible tubular member comprises an inner cross-sectional surface profile that matches an outer cross-sectional surface profile of the distal segment of the inner tubular layer. [0094] Aspect 12. The expandable introducer apparatus of any one of Aspects 10-11, wherein the flexible tubular member comprises an inner circumferential flared segment comprising a flared end abutting a folded end of the outer fold of the foldable region. [0095] Aspect 13. The expandable introducer apparatus of Aspect 10, wherein the flexible tubular member comprises a first inner protrusion aligned with a folded end of the outer fold of the foldable region and a second inner protrusion aligned with a folded end of the inner fold of the foldable region. [0096] Aspect 14. The expandable introducer apparatus of any one of Aspects 1-10, wherein the flexible tubular member further comprises a circular cross-sectional area with a uniform wall thickness circumscribing the distal segment of the inner tubular layer. [0097] Aspect 15. The expandable introducer apparatus of any one of Aspects 1-14, wherein the outer tubular layer comprises an elongated split extending along the radially expandable region in a direction of the elongated axis. [0098] Aspect 16. A method of guiding an object through an inner lumen of an expandable introducer apparatus, the expandable introducer apparatus comprising: an inner tubular layer comprising the inner lumen and a radially expandable region extending along an elongated axis of the inner lumen; an outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer; and a flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer, the flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment, and the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member, the method comprising distally advancing the object through a proximal end of the inner lumen, wherein a portion of the radially expandable region radially expands from a contracted orientation to an expanded orientation while a corresponding portion of the outer tubular layer expands to accommodate the expanded orientation of the radially expandable region. The method further comprising distally advancing the object past the distal end of the outer tubular layer to radially expand the radially expandable region of the inner tubular layer along the axial length of the distal segment while simultaneously circumferentially stretching the flexible region of the flexible tubular member, wherein the distal segment of the inner tubular layer and the flexible tubular member simultaneously expand to accommodate the object passing through the distal segment of the inner tubular layer. Distally advancing the object past a distal end of the inner tubular layer. [0099] Aspect 17. The method of Aspect 16, wherein a distal end of the flexible tubular member extends distally beyond the distal end of the inner tubular layer to form a flexible tip, and the method further comprises distally advancing the object through the flexible tip and past the distal end of the flexible tubular member. [0100] Aspect 18. The method of Aspects 17, wherein the flexible tip forms a rounded transition at the distal end of the inner tubular layer when advancing the object past the distal end of the inner tubular layer and through the flexible tip. [0101] Aspect 19. The method of any one of Aspects 17-18, wherein the distal end of the flexible tubular member forms a circumferential seal against an outer circumferential surface of the object as the object passes through the flexible tip and past the distal end of the flexible tubular member. [0102] Aspect 20. The method of any one of Aspects 17-19, wherein a length of the flexible tip from the distal end of the inner tubular layer to the distal end of the flexible tubular member is from about 0.5 mm to about 3 mm. [0103] Aspect 21. The method of any one of Aspects 16-20, further comprising proximally retracting the object within the inner lumen proximally past the distal end of the outer

tubular layer and out of the inner lumen of the distal segment of the inner tubular layer, wherein the flexible tubular member contracts to contract the radially expandable region of the distal segment of the inner tubular layer. [0104] Aspect 22. The method of Aspect 21, wherein the radially expandable region of the distal segment of the inner tubular layer comprises a foldable region, wherein radially contracting the flexible tubular member at least partially folds the foldable region. [0105] Aspect 23. The method of any one of Aspects 16-21, wherein the radially expandable region comprises a foldable region, wherein radially expanding the radially expandable region at least partially unfolds the foldable region. [0106] Aspect 24. The method of any one of Aspects 16-23, wherein the outer tubular layer comprises an elongated split extending along the radially expandable region in a direction of the elongated axis, and the distally advancing the object through the proximal end of the inner lumen radially expands the portion of the radially expandable region while a corresponding portion of the elongated split separates. [0107] It should be understood that while various aspects have been described in detail relative to certain illustrative and specific examples thereof, the present disclosure should not be considered limited to such, as numerous modifications and combinations of the disclosed features are possible without departing from the scope of the following claims.

Claims

1. An expandable introducer apparatus comprising: an inner tubular layer comprising an inner lumen and a radially expandable region extending along an elongated axis of the inner lumen and configured to at least partially radially expand relative to the elongated axis from a contracted orientation to an expanded orientation; an outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer; and a flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer, the flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment, and the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member.
2. The expandable introducer apparatus of claim 1, wherein the flexible tubular member comprises a proximal end, a distal end, and a length extending from the proximal end to the distal end of the flexible tubular member in the direction of the elongated axis.
3. The expandable introducer apparatus of claim 2, wherein the length is from about 5 mm to about 7 mm.
4. The expandable introducer apparatus of claim 2, wherein the proximal end of the flexible tubular member abuts the distal end of the outer tubular layer.
5. The expandable introducer apparatus of claim 2, wherein the flexible tubular member comprises a flexible tip that extends beyond a distal end of the inner tubular layer.
6. The expandable introducer apparatus of claim 5, wherein the flexible tip comprises a length extending from the distal end of the inner tubular layer to the distal end of the flexible tubular member.
7. The expandable introducer apparatus of claim 6, wherein the length of the flexible tip is from about 0.5 mm to about 3 mm.
8. The expandable introducer apparatus of claim 7, wherein the length of the flexible tip is from about 1 mm to about 1.5 mm.
9. The expandable introducer apparatus of claim 1, wherein the radially expandable region of the inner tubular layer comprises a foldable region configured to be at least partially unfold from the contracted orientation to the expanded orientation.
10. The expandable introducer apparatus of claim 9, wherein the foldable region further comprises

an outer fold and an inner fold in the contracted orientation.

11. The expandable introducer apparatus of claim 10, wherein the flexible tubular member comprises an inner circumferential flared segment comprising a flared end abutting a folded end of the outer fold of the foldable region.

12. The expandable introducer apparatus of claim 10, wherein the flexible tubular member comprises a first inner protrusion aligned with a folded end of the outer fold of the foldable region and a second inner protrusion aligned with a folded end of the inner fold of the foldable region.

13. The expandable introducer apparatus of claim 1, wherein the flexible tubular member comprises an inner cross-sectional surface profile that matches an outer cross-sectional surface profile of the distal segment of the inner tubular layer.

14. The expandable introducer apparatus of claim 1, wherein the flexible tubular member further comprises a circular cross-sectional area with a uniform wall thickness circumscribing the distal segment of the inner tubular layer.

15. A method of guiding an object through an inner lumen of an expandable introducer apparatus, the expandable introducer apparatus comprising: an inner tubular layer comprising the inner lumen and a radially expandable region extending along an elongated axis of the inner lumen; an outer tubular layer comprising an inner surface bonded to the inner tubular layer, wherein a distal segment of the inner tubular layer extends beyond a distal end of the outer tubular layer; and a flexible tubular member comprising a bonded region and a flexible region, wherein an inner surface of the bonded region is bonded to an outer surface of the distal segment of the inner tubular layer, the flexible region is not bonded to the radially expandable region of the inner tubular layer along an axial length of the distal segment, and the radially expandable region of the inner tubular layer along the axial length of the distal segment extends under the flexible region of the flexible tubular member, the method comprising: distally advancing the object through a proximal end of the inner lumen, wherein a portion of the radially expandable region radially expands from a contracted orientation to an expanded orientation while a corresponding portion of the outer tubular layer expands to accommodate the expanded orientation of the radially expandable region; distally advancing the object past the distal end of the outer tubular layer to radially expand the radially expandable region of the inner tubular layer along the axial length of the distal segment while simultaneously circumferentially stretching the flexible region of the flexible tubular member, wherein the distal segment of the inner tubular layer and the flexible tubular member simultaneously expand to accommodate the object passing through the distal segment of the inner tubular layer; and distally advancing the object past a distal end of the inner tubular layer.

16. The method of claim 15, wherein a distal end of the flexible tubular member extends distally beyond the distal end of the inner tubular layer to form a flexible tip, and the method further comprises distally advancing the object through the flexible tip and past the distal end of the flexible tubular member.

17. The method of claim 16, wherein the flexible tip forms a rounded transition at the distal end of the inner tubular layer when advancing the object past the distal end of the inner tubular layer and through the flexible tip.

18. The method of claim 16, wherein the distal end of the flexible tubular member forms a circumferential seal against an outer circumferential surface of the object as the object passes through the flexible tip and past the distal end of the flexible tubular member.

19. The method of claim 18, wherein a length of the flexible tip from the distal end of the inner tubular layer to the distal end of the flexible tubular member is from about 0.5 mm to about 3 mm.

20. The method of claim 15, wherein the radially expandable region comprises a foldable region, wherein radially expanding the radially expandable region at least partially unfolds the foldable region.
